Fusion Constructive Realism (FCR): Complete Mathematical Specification and Algorithmic Framework

Summary Document for State Transfer & Context Persistence

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Abstract

This document provides a fully self-contained mathematical, physical, and computational summary of Fusion Constructive Realism (FCR). FCR is a non-local, constructive, and finite relational network framework in which spacetime geometry and particle physics emerge from discrete correlation state spaces ($\mathcal{M}_n$). We outline the core definitions, local mutation rules, phase venting/fission mechanics, path interference principles, boundary identification schemas, canonical deduplication algorithms, and repository state storage specifications.

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1 Foundational Philosophy and State Space Representation

Fusion Constructive Realism rejects pre-existing continuum spacetime and local spatial realism. Instead, physical reality is modeled as a discrete relational graph $Q = (P, \mu, C)$ at poset step T.

1.1 State Representation

A state $Q \in \mathcal{M}_n$ is defined by a triad (P, μ, C):

1. **Base Set (P):** A finite set of n discrete node identifiers $P = \{0, 1, \ldots, n-1\}$.
2. **Correlation Field (C):** A matrix of complex values $z_{ij} \in \mathbb{D}_{\mathbb{Q}(i)}$ representing quantum phase and magnitude coupling between nodes $i, j \in P$, subject to:

$$|C(i,j)| \leq 1.0, \quad C(i,i) = 1.0 + 0.0i, \quad C(i,j) = C(j,i)^* \tag{1}$$

3. **Primitive Metric (μ):** A matrix of spatial distance values $\mu(i,j) \in \mathbb{Q}^+$:

$$\mu(i,j) = \begin{cases} 1.0 \text{ (baseline rest metric)}, & \text{if } C(i,j) \neq 0 \\ \text{ShortestPath}_\mu(i \rightsquigarrow j), & \text{if } C(i,j) = 0 \\ \infty, & \text{if disjoint/unconnected} \end{cases} \tag{2}$$

with self-distance $\mu(i,i) = 0.0$.

1.2 Node Capacity Bound

Physical stability requires that no single node can support unbounded external correlations. For every node $p \in P$, the external capacity sum $S(p)$ must satisfy:

$$S(p) = \sum_{q \neq p} |C(p,q)|^2 \leq 1.0 \tag{3}$$

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2 Targeted Local Mutation Alphabet (Σ)

State evolution across discrete poset time $T \to T+1$ is governed by local operational mutations from the alphabet $\Sigma = \{t, d, \mu, v\}$.

2.1 1. Triangulation (t): Structural Closure

Triangulation acts strictly as a **one-time topological closure rule** on missing direct edges ($C(i,j) = 0$). Given an open 2-path $i \to x \to j$:

$$C(i,j) \leftarrow C(i,x) \cdot C(x,j) \quad \text{iff } C(i,j) = 0 \tag{4}$$

Once an edge exists ($C(i,j) \neq 0$), Triangulation no longer acts on that specific edge.

2.2 2. Correlation Decay (d)

Correlation magnitudes dissipate over spatial distance according to:

$$C_{T+1}(i,j) = C_T(i,j) \cdot e^{-\gamma_d \mu_T(i,j)} \tag{5}$$

where $\gamma_d > 0$ is the decay constant. If $|C(i,j)| < \epsilon$ ($\epsilon = 10^{-6}$), $C(i,j) \to 0$ (nilpotent link severing).

2.3 3. Metric Evolution (μ)

Spatial metric distance contracts under dense correlation density and relaxes toward baseline rest distance $d_{\text{rest}} = 1.0$:

$$\mu_{T+1}(i,j) = \mu_T(i,j) - \alpha|C_T(i,j)|^2 + \beta(1.0 - \mu_T(i,j)) \tag{6}$$

where α is the contraction coupling and β is the spatial elasticity recovery rate.

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3 Capacity Normalization, Phase Venting ($\mathcal{M}_{1/2}$), and Fission

When a mutation pushes node p past unit capacity ($S(p) > 1.0$), the system executes dynamic normalization and topological fission:

3.1 Phase Venting ($\mathcal{M}_{1/2}$ Virtual Half-Edge)

The node sheds an excess capacity magnitude $\Delta S = S(p) - 1.0$. The phase angle ϕ_{vent} of the emitted $\mathcal{M}_{1/2}$ virtual half-edge conserves net phase momentum:

$$\phi_{\text{vent}} = \arg\left(\sum_{q\neq p} C(p,q)\right) \tag{7}$$

3.2 Coherent Amplitude Scaling

All active edge correlations connected to overloaded node p are projected back to unity without altering relative phase orientations:

$$C(p,q) \leftarrow \frac{C(p,q)}{\sqrt{S(p)}} \implies \sum_{q\neq p}\left|\frac{C(p,q)}{\sqrt{S(p)}}\right|^2 = 1.0 \tag{8}$$

3.3 Link Severing & Topological Fission

Any edge magnitude falling below noise threshold $|C(p,q)| < \epsilon$ is severed ($C(p,q) \to 0$). Connected-component analysis is performed on P:

- The primary connected component Q_1 continues as the core attractor at level n.
- Severed subgraphs $Q_2, \ldots$ are ejected as radiation remnants ($\mathcal{M}_2$ chains).

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4 Path Superposition, Interference, and Class-2 Attractors

4.1 Path Superposition

When multiple parallel paths $\gamma_1, \gamma_2, \ldots$ exist between nodes p and q, correlation amplitudes add constructively or destructively via complex path sums:

$$C_{\text{net}}(p,q) = \sum_k Z(\gamma_k) = \sum_k\left(\prod_{e\in\gamma_k} z_e\right) = \sum_k R_k e^{i\Phi_k} \tag{9}$$

4.2 Interference Conditions

For two convergent paths with amplitudes $Z_1 = R_1 e^{i\Phi_1}$ and $Z_2 = R_2 e^{i\Phi_2}$:

$$|Z_{\text{net}}|^2 = R_1^2 + R_2^2 + 2R_1R_2\cos(\Phi_1 - \Phi_2) \tag{10}$$

- **Constructive Interference ($\Delta\Phi = 2m\pi$):** $|Z_{\text{net}}| \to R_1 + R_2$. Amplifies correlation, accelerating metric contraction ($\mu \to 0$).
- **Destructive Interference ($\Delta\Phi = (2m+1)\pi$):** $|Z_{\text{net}}| \to |R_1 - R_2|$. Attenuates correlation, triggering link decay and fission.

4.3 Mechanism of Class-2 Stable Attractors

A closed topology (such as the $n = 3$ triangle $\mathcal{M}_3$ or $n = 6$ octahedron $\mathcal{M}_6$) achieves non-decaying limit cycle stability because metric contraction ($\mu \to 0$) cancels spatial decay ($e^{-\gamma_d \mu} \to 1.0$). At equilibrium ($r^* = \mu^*$):

$$|C_{T+1}| = |C_T| > 0, \quad \mu_{T+1} = \mu_T > 0 \tag{11}$$

Phase rotates indefinitely around closed loops without magnitude loss or capacity blowup.

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5 Attractor Boundary Assembly Schema ($\mathcal{M}_n^{\text{seed}}$)

To generate target topologies without combinatorial state explosion, initial seeds at $T = 0$ are constructed by gluing lower-order attractors ($n' < n$) along explicit boundary maps:

$$\mathcal{M}_n^{\text{seed}} = \left(\bigotimes_k A_k,\ B,\ \sigma \right) \tag{12}$$

5.1 Example: Octahedral Shell ($\mathcal{M}_3^{\wedge 4} \to \mathcal{M}_6$)

To construct the 6-node Octahedron ($\mathcal{M}_6^{\text{oct}}$), 4 triangles ($\mathcal{M}_3$) are glued along a 2-to-1 vertex identification map B, yielding 6 global nodes with valence 4:

$$\begin{aligned}
&\text{Global Node } 0 \equiv T_A^{(0)} \sim T_B^{(0)}, && \text{Global Node } 3 \equiv T_B^{(1)} \sim T_C^{(1)} \\
&\text{Global Node } 1 \equiv T_A^{(1)} \sim T_C^{(0)}, && \text{Global Node } 4 \equiv T_B^{(2)} \sim T_D^{(1)} \\
&\text{Global Node } 2 \equiv T_A^{(2)} \sim T_D^{(0)}, && \text{Global Node } 5 \equiv T_C^{(2)} \sim T_D^{(2)}
\end{aligned}$$

Under Mutation t, phase product conduction across open 2-paths automatically fills the 4 remaining faces, completing all 8 faces of the $\mathcal{M}_6^{\text{oct}}$ shell.

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6 Canonical Reduction and Deduplication Pipeline

To eliminate automorphism redundancy (where node permutations, phase rotations, and float drift create duplicate state entries), all candidate states pass through a 3-step reduction function:

1. **Global Phase Gauge-Fixing:** Rotate all edge phases relative to the first non-zero edge $C(i, j)$:
$$C_{\text{fixed}}(a, b) = C(a, b) \cdot e^{-i\arg(C(i,j))} \implies \operatorname{Im}(C_{\text{fixed}}(i, j)) = 0 \tag{13}$$

2. **Numerical Binning:** Round floating-point values for $|z|$, $\arg(z)$, and μ to fixed precision $\epsilon_{\text{bin}} = 10^{-3}$ to collapse local floating-point drift.

3. **Isomorphism Canonical Hashing:** Pass the binned matrix to the Weisfeiler-Lehman / VF2 graph hash algorithm (‘nx.weisfeiler$_{l}eman_{g}raph_{h}ash$‘$), returningauniquedeterministicStringHash$

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7 Repository Storage and Probability Calculation

States are stored on disk in the path-aware directory hierarchy: `fcr_repository/M<n>/T_<k>/state_<CANONICA`

7.1 Born-Rule Probability Analyzer (`fcr_analyzer.py`)

The total display probability $P(Q)$ for reaching state Q at timestep T_k is computed across all convergent path trajectories $\gamma \to Q$:

$$P(Q) = \frac{\left|\sum_{\gamma \to Q} Z(\gamma)\right|^2}{\sum_{Q' \in T_k} \left|\sum_{\gamma' \to Q'} Z(\gamma')\right|^2} \tag{14}$$

where $Z(\gamma) = \prod_{e \in \gamma} z_e$ is the cumulative complex phase product along trajectory γ.

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8 Current Development Status and Roadmap

Module / Branch	**Status**	**Core Functionality**
Branch `INITIAL_REPL`	Committed	Interactive REPL (`fcr_repl.py`), basic mutation rules
Theoretical Formalism	Complete	Exact capacity scaling, fission thresholds, boundary gluing
Canonical Reduction Layer	Specified	Gauge-fixing, float binning, Weisfeiler-Lehman deduplication
Attractor Search Engine	Active	DFS parameter calibration ($\alpha \geq 2\beta$) for $n = 3, 4, 6$
Path Probability Engine	Specified	Trajectory graph assembly, $P(d_k)$ Born-rule calculation

Table 1: System Status Overview.